

Simone Maccario

MSc Computer Science Student, ETH Zurich | Data & Software

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Portfolio | LinkedIn | GitHub

PROFILE

MSc Computer Science student at ETH Zurich (BSc in Informatics, USI) with hands-on experience in Python data pipelines on AWS, static analysis at scale (100k+ code executions) and full-stack development (Spring Boot, React, Vue.js, PostgreSQL, Docker). Reliable, fast learner and team player shaped by competitive football. Open to internships and part-time roles in software and data engineering.

EDUCATION

MSc in Computer Science – Data Management & Machine Intelligence *Sep 2026 – Present*
ETH Zurich, Switzerland

BSc in Informatics – magna cum laude *Sep 2023 – Jun 2026*
Università della Svizzera italiana (USI), Lugano, Switzerland

- **Thesis:** *From Raw Data to Pedagogical Insights: An Interactive Dashboard Leveraging Static Analysis for PyTamaro-Web* (supervisor: Prof. Matthias Hauswirth) | Thesis PDF
- Built a Python pipeline fusing PostgreSQL telemetry and Jupyter Notebooks to rebuild user sessions.
- Scaled static analysis (ty type checker) to 106,080 executions via batching and a global cache, skipping ∼29,000 redundant runs and cutting full-dataset processing from minutes per variant to ∼76 s.
- Generated an offline dashboard (Jinja2, Tailwind CSS, Monaco Editor) with execution timelines, code diffs and error highlighting; validated in a usability study with an instructor.

Scientific High School (Liceo Scientifico Scienze Applicate), Liceo Paolo Giovio, Como, Italy *2018 – 2023*

PROFESSIONAL EXPERIENCE

Data Engineering Intern *Sep 2025 – Dec 2025*
Brightside Capital, Lugano, Switzerland

- Built Python automation to import, map and export banking data from AWS into the internal database schema.
- Mapped heterogeneous banking datasets to a unified schema and built validation checks to ensure data reliability.
- Diagnosed and fixed existing issues in the data-processing pipeline and contributed to its ongoing development.

SELECTED PROJECTS

Frantic: Pandora's Box – Digital Companion App (personal, solo) *Live demo | GitHub*
React, TypeScript, Vite, Tailwind CSS, Vitest, Netlify

- Mobile-first scorekeeper and rules companion for the card game Frantic: automated scoring, live leaderboard, rule-card randomiser and a searchable 94-card encyclopedia in 4 languages (EN, DE, FR, IT).
- Typed game-state engine with auto-save and JSON import/export; 20 Vitest test suites; hosted on Netlify.

OptiMove – SAT-Based Logistics Planner (team project, frontend lead) *GitHub*
Python, PySAT (Glucose-3), FastAPI, Vue 3, Tailwind CSS, Docker

- Encoded multi-agent logistics planning as propositional CNF (13 constraint groups) to find time-optimal plans; FastAPI backend with process-isolated solver runs and an animated Vue 3 visualiser.

Meme Search Engine (team of 2, backend lead) *GitHub*
Python, Scrapy, Playwright, Apache Solr 9, Carrot2, Vue 3, Nginx, Docker Compose

- Containerised IR system: crawlers for 4 platforms with deduplication and caching, automated Solr indexing and Carrot2 Lingo topic clustering, served to a Vue 3 UI via an Nginx reverse proxy.

SEWorld – AI-Augmented Educational Platform (team of ∼20) *GitHub*
Java 21, Spring Boot 3, Spring AI, Ollama, PostgreSQL 17, Vue 3, Docker

- Built frontend and backend features for LLM-based exercise evaluation, student statistics and dynamic feedback.

Further projects: Battleship (Java, design patterns, game AI) • Snake (Racket, 140 unit tests) • Portfolio (Vue 3, GitHub Actions)

SKILLS & INTERESTS

Programming	Python, Java, TypeScript, JavaScript, SQL, C, C++, Bash, Racket
Data & Backend	PostgreSQL (schema design, CTEs, MySQL migration), Solr, Spring Boot, FastAPI, REST
Frontend	React, Vue 3, Vite, Tailwind CSS, Bootstrap, Jinja2, HTML5, CSS3, Web Components
DevOps & Testing	AWS, Docker, Nginx, GitHub Actions, Netlify, Git, GitLab, Linux, Vitest, JUnit, pytest, LaTeX
Languages	Italian (native) • English (B2)
Interests	Former professional-level football player • Motorbiking • Skiing • Film and TV